Web Application Development Lab 04 Report

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Exercise 1

Task 1.2

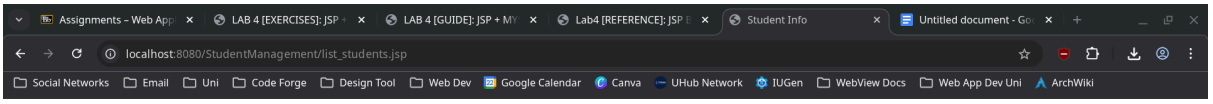
Code

```
<%--
    Document   : list_students
    Created on : Apr 22, 2026, 2:16:39 PM
    Author     : islabre
--%>

<%@page contentType="text/html" pageEncoding="UTF-8"%>
<%@ page import="java.sql.*" %>
<!DOCTYPE html>
<html>
<head>
    <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
    <title>Student Info</title>

    <style>
        table, td {
            border: 1px solid;
            border-collapse: collapse;
        }

        td {
            padding: 0.25em;
        }
    </style>
</head>
<body>
    <h1>Student Info</h1>

    <% if (request.getParameter("message") != null) { %>
        <div class="message">
            <%= request.getParameter("message") %>
```

```
    </div>
<% } %>

<table>
    <thead>
        <td>ID</td>
        <td>Student Code</td>
        <td>Full Name</td>
        <td>Email</td>
        <td>Major</td>
        <td>Created At</td>
    </thead>
    <tbody>
    <%
        Connection conn = null;
        Statement stmt = null;
        ResultSet rs = null;

        try {
            // Connect to database
            Class.forName("com.mysql.cj.jdbc.Driver");

            conn = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/student_management",
                "root",
                "Q@jaP*96T8xY#2^7&3@7"
            );

            // Query for students
            stmt = conn.createStatement();
            String sql = "SELECT * FROM students";
            rs = stmt.executeQuery(sql);

            // Get info
            while (rs.next()) {
                int id = rs.getInt("id");
                String studentCode = rs.getString("student_code");
                String fullName = rs.getString("full_name");
                String email = rs.getString("email");
                String major = rs.getString("major");
                Timestamp createdAt = rs.getTimestamp("created_at");
    %>
                <tr>
                    <td><%= id %></td>
                    <td><%= studentCode %></td>
                    <td><%= fullName %></td>
                    <td><%= email != null ? email : "N/A" %></td>
                    <td><%= major != null ? major : "N/A" %></td>
```

```
                <td><%= createdAt %></td>
            </tr>
        <%
            }
        // Handle errors
        } catch (ClassNotFoundException e) {
            out.println("<tr><td colspan='7'>Error: JDBC Driver not found!</td></tr>");
            e.printStackTrace();
        } catch (SQLException e) {
            out.println("<tr><td colspan='7'>Database Error: " + e.getMessage() +
"</td></tr>");
            e.printStackTrace();
        // Close database connection
        } finally {
            try {
                if (rs != null) rs.close();
                if (stmt != null) stmt.close();
                if (conn != null) conn.close();
            } catch (SQLException e) {
                e.printStackTrace();
            }
        }
    %>
    </tbody>
  </table>
</body>
</html>
```

Result



ID	Student Code	Full Name	Email	Major	Created At
1	SV001	John Smith	john.smith@email.com	Computer Science	2026-04-22 07:18:54.0
2	SV002	Emily Johnson	emily.j@email.com	Information Technology	2026-04-22 07:18:54.0
3	SV003	Michael Brown	michael.b@email.com	Software Engineering	2026-04-22 07:18:54.0
4	SV004	Sarah Davis	sarah.d@email.com	Data Science	2026-04-22 07:18:54.0
5	SV005	David Wilson	david.w@email.com	Computer Science	2026-04-22 07:18:54.0

Exercise 2

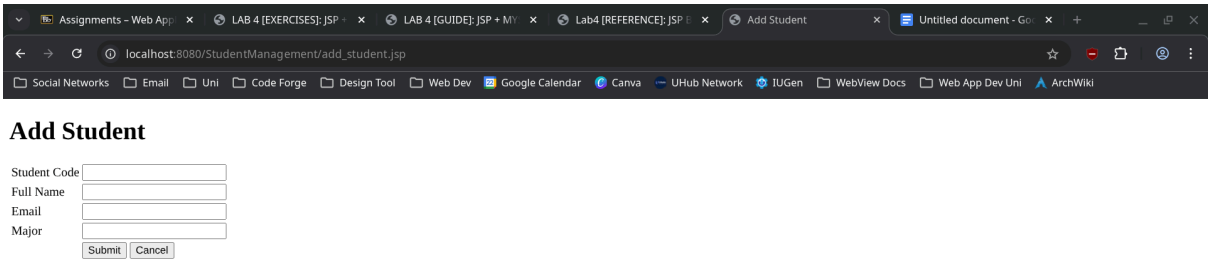
Task 2.1

Code

```
<%--
   Document   : process_add
   Created on : Apr 22, 2026, 2:56:45 PM
   Author     : islabre
--%>

<%@page contentType="text/html" pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<head>
   <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
   <title>Add Student</title>
</head>
<body>
   <h1>Add Student</h1>

   <% if (request.getParameter("message") != null) { %>
      <div class="message">
         <%= request.getParameter("message") %>
      </div>
   <% } %>

   <form method="POST" action="process_add.jsp">
      <table>
         <tr>
            <td>Student Code</td>
            <td>
               <input type="text" name="student-code" id="student-code" required>
            </td>
         </tr>
         <tr>
            <td>Full Name</td>
            <td>
               <input type="text" name="full-name" id="full-name" required>
            </td>
         </tr>
         <tr>
            <td>Email</td>
            <td>
               <input type="email" name="email" id="email">
```

```
            </td>
        </tr>
        <tr>
            <td>Major</td>
            <td>
                <input type="text" name="major" id="major">
            </td>
        </tr>
        <tr>
            <td></td>
            <td>
                <input type="submit" value="Submit">
                <a href="list_students.jsp">
                    <button type="button">Cancel</button>
                </a>
            </td>
        </tr>
    </table>
  </form>
</body>
</html>
```

Result



Task 2.2

Code

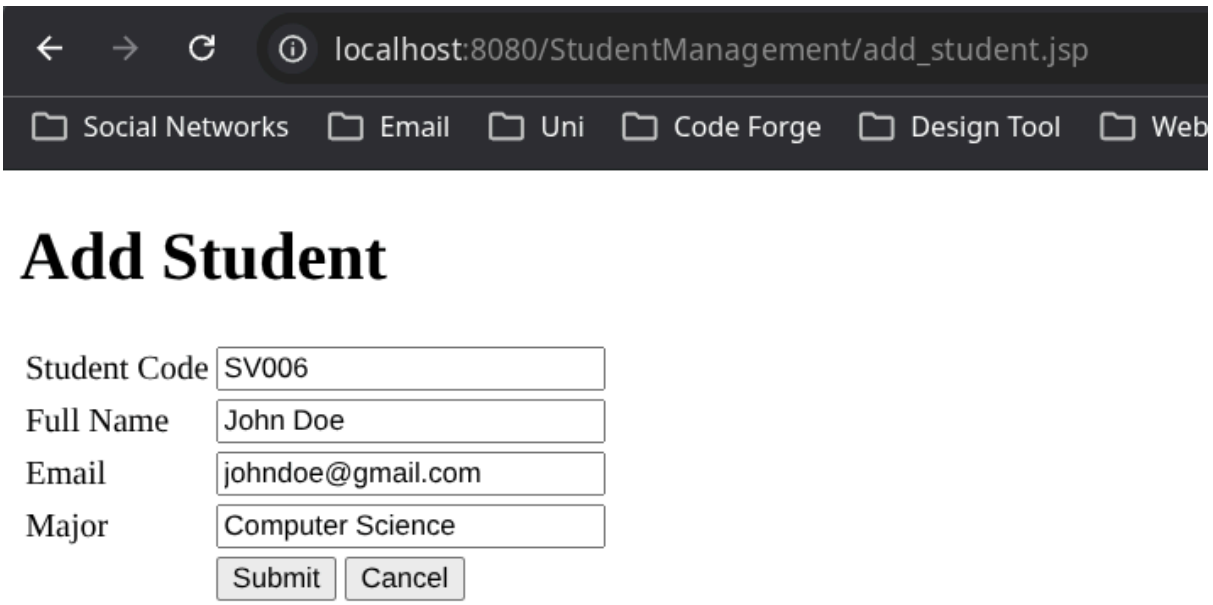
```
<%--
```

```
    Document   : process_add
    Created on : Apr 22, 2026, 3:13:22 PM
    Author     : islabre
--%>

<%@page contentType="text/html" pageEncoding="UTF-8"%>
<%@ page import="java.sql.*" %>
<!DOCTYPE html>
<html>
<head>
    <meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
    <title>Student Adding Process</title>
</head>
<body>
<%
    String studentCode = request.getParameter("student-code");
    String fullName = request.getParameter("full-name");
    String email = request.getParameter("email");
    String major = request.getParameter("major");

    if (studentCode == null || studentCode.trim().isEmpty() ||
        fullName == null || fullName.trim().isEmpty()) {
        response.sendRedirect("add_student.jsp?message=Error: Required fields are
missing.");
    }

    Connection conn = null;
    PreparedStatement pstmt = null;

    try {
        // Connect to database
        Class.forName("com.mysql.cj.jdbc.Driver");

        conn = DriverManager.getConnection(
            "jdbc:mysql://localhost:3306/student_management",
            "root",
            "Q@jaP*96T8xY#2^7&3@7"
        );

        // Insert info into database (prepare statement first to prevent SQL Injection Attack)
        String sql = "INSERT INTO students (student_code, full_name, email, major) VALUES
(?, ?, ?, ?)";

        pstmt = conn.prepareStatement(sql);
        pstmt.setString(1, studentCode);
        pstmt.setString(2, fullName);
        pstmt.setString(3, email);
        pstmt.setString(4, major);
```

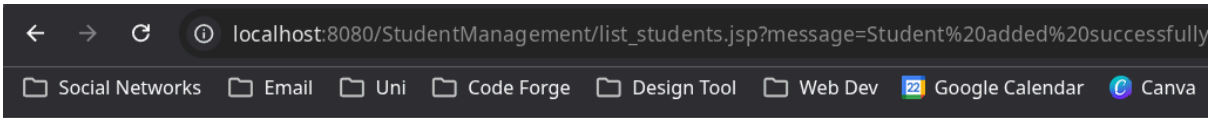
```
        int rowsAffected = pstmt.executeUpdate();

        if (rowsAffected > 0) {
            response.sendRedirect("list_students.jsp?message=Student added successfully");
        } else {
            response.sendRedirect("add_student.jsp?message=Error: Failed to add student");
        }

        // Handle errors
        } catch (ClassNotFoundException e) {
            response.sendRedirect("add_student.jsp?message=Error: JDBC Driver not found.");
            e.printStackTrace();
        } catch (SQLException e) {
            String errorMsg = e.getMessage();
            if (errorMsg.contains("Duplicate entry")) {
                response.sendRedirect("add_student.jsp?message=Error: Student code already
exists");
            } else {
                response.sendRedirect("add_student.jsp?message=Error: Database error");
            }
            e.printStackTrace();

        // Close database connection
        } finally {
            try {
                if (pstmt != null) pstmt.close();
                if (conn != null) conn.close();
            } catch (SQLException e) {
                e.printStackTrace();
            }
        }
%>
</body>
</html>
```

Result

Adding a student via add_student.jsp:



After adding:

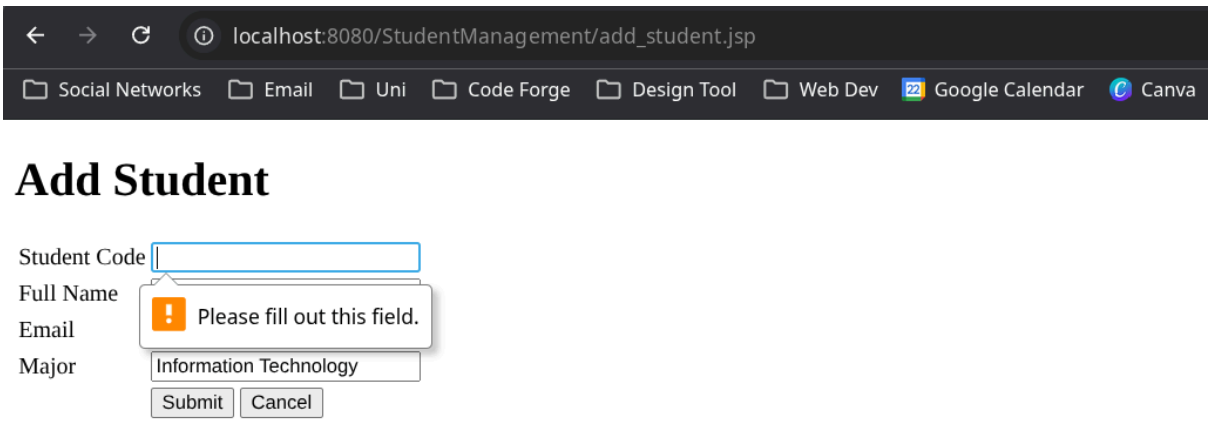


Student Info

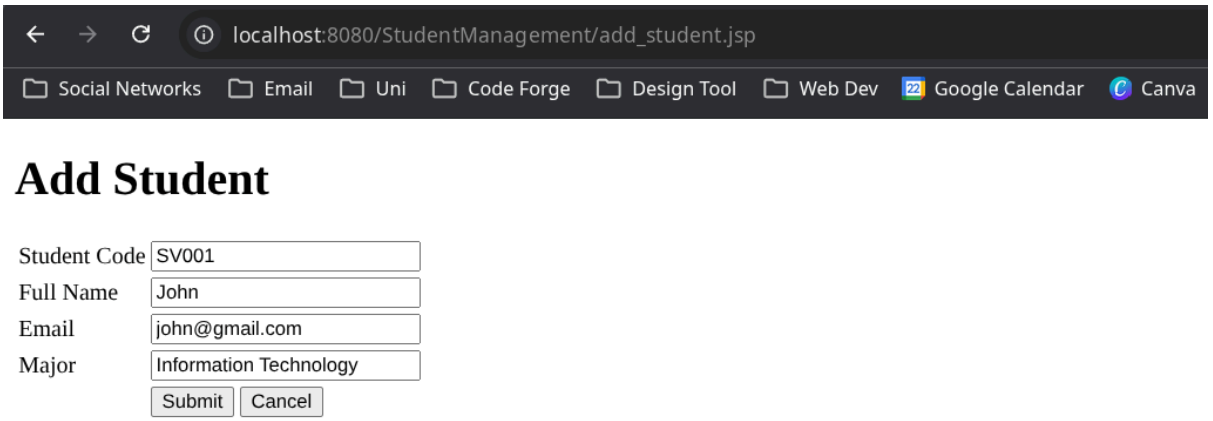
Student added successfully

ID	Student Code	Full Name	Email	Major	Created At
1	SV001	John Smith	john.smith@email.com	Computer Science	2026-04-22 07:18:54.0
2	SV002	Emily Johnson	emily.j@email.com	Information Technology	2026-04-22 07:18:54.0
3	SV003	Michael Brown	michael.b@email.com	Software Engineering	2026-04-22 07:18:54.0
4	SV004	Sarah Davis	sarah.d@email.com	Data Science	2026-04-22 07:18:54.0
5	SV005	David Wilson	david.w@email.com	Computer Science	2026-04-22 07:18:54.0
7	SV006	John Doe	johndoe@gmail.com	Computer Science	2026-04-22 08:36:26.0

Try to add a student with empty Student Code (handled by the browser itself):



Try to add a student using existing Student Code:



Result:

localhost:8080/StudentManagement/add_student.jsp?message=Error:%20Student%20code%20alre

Social Networks | Email | Uni | Code Forge | Design Tool | Web Dev | Google Calendar | Canva

Add Student

Error: Student code already exists

Student Code

Full Name

Email

Major

Submit Cancel